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Technical Skills

- **Programming:** Matlab, Python, R, C++, Cuda, Bash
- **Deep Learning:** CNN, RNN, GAN
- **Machine Learning:** Supervised, unsupervised, semi-supervised and self-supervised methods
- **Image Processing:** Image segmentation, denoising, super-resolution reconstruction and object detection
- **Database:** SQL
- **Méthodology:** Agile
- **MRI:** MR images acquisition, functional and anatomical MR images
- **Tools:** Docker, 3D slicer, Git, Visual Studio

Certifications

- **Udacity:** Generative IA
- **Nvidia Deep Learning Institute :** The fundamental of Deep Learning
- **Coursera:** Deep Learning Specialization

Languages

- **Sara:** Mother tongue
- **French:** Working language
- **English:** Working language
- **Swahili:** Beginner
- **Wolof:** Beginner

Soft Skills

- Team work attitude and good communication
- Adaptation et self-learning

Interests

- Reading, sport and travel

Data Scientist - Machine Learning Engineer

Experiences

June 2025-Now	Postdoctoral Scholar	IADI lab-CHRU Nancy
• MR images acquisition and estimation of MR relaxation time T_1 et T_2 using the new advanced PLANET model (bssfp sequence) • Technical stack: Matlab, Bash		
October 2021-May 2025	Doctoral Scholar	IADI lab-CHRU Nancy
• Human ex-vivo placental MR images acquisition • Algorithms development for placental super-resolved MR images, the estimation of MR biomarkers (perfusion, diffusion) and ex-vivo placental vessels segmentation • Teaching: Diffusion MRI course • Scientific publications and presentation in international conferences • Technical stack: Matlab, Python (Pytorch), 3D slicer, Bash, Git, Docker		
March 2021-September 2021	Fatima Fellowship Program (Mentoring)	Remote
• Mentoring program (research), mentor: Marouen Ben Guebila (Harvard University, USA) • Project: Bayesian Optimization for Non-convex Space Sampling		
June 2020 - October 2020	Data scientist Intern	Occitanie Data, Toulouse, France
• Exploration of federated learning for images classification and homomorphic encryption for regression problem • Technical stack: Python (PySift)		

Education

2025	Université de Lorraine Ph.D. in Medical Imaging	Nancy, France
2019	Sorbonne Université (Université Pierre et Marie Curie) Master in Mathematical Modeling	Paris, France
2018	Aix Marseille Université Master in Mathematical Engineering	Marseille, France
2015	Université Cheikh Anta Diop Bachelor in Mathematics	Dakar, Senegal

Projects

2025	Early detection of gynecological malignancies using ensemble deep learning models Stack technique: Python, Pytorch
2024	Lung segmentation using CT images Image modality: CT images. Learning model: nn-Unet. Technical stack: Python, Pytorch

